- *Cloud Trace (https://cloud.google.com/trace)*—Google's offering for distributed tracing, which supports Go, Java, Node.js, and Python via OpenTelemetry.

Figure 5.4 provides an example output of Google Cloud Trace, a sophisticated distributed tracing backend. It shows the analytics of traces, specifically the latency distribution. Next, we will cover open source tracing solutions.

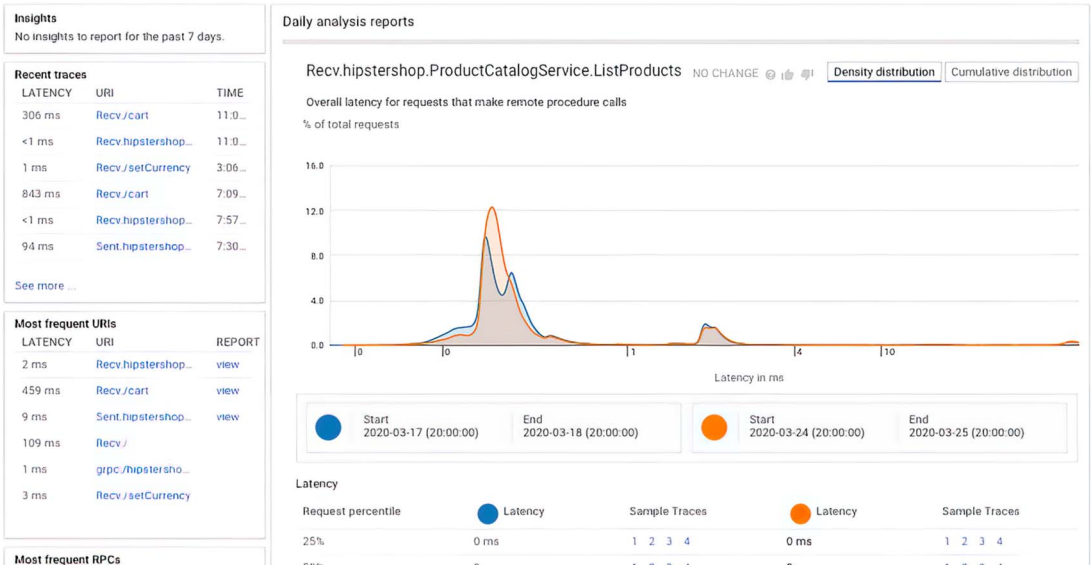



**Figure 5.4 Viewing traces in Google Cloud Trace (Google Cloud docs; https://cloud.google.com/trace/docs/trace-overview)**

### 5.4.2 Open source traces backends

Open source tracing solutions do not offer as many options as with metrics backends, but they still offer some choice:

- *CNCF Jaeger and Zipkin*—Both Jaeger (https://github.com/jaegertracing/jaeger) and Zipkin (https://github.com/openzipkin/zipkin) are distributed tracing systems hosted by CNCF. These days, Jaeger seems to be the more actively developed project, but Zipkin is still widely used and supported. I recommend you have a look at the respective communities (Slack and GitHub issues) and make up your own mind. We will cover both in chapter 6.
- *Elasticsearch and OpenSearch*—These options have expanded from logs to support handling traces. Both have excellent frontends (Kibana and Dashboards) that we will, again, cover in chapter 6.
- *Grafana Tempo (https://grafana.com/oss/tempo/)*—A distributed tracing backend available under the AGPLv3 license. It uses object storage (e.g., S3) in the